



Universidade do Minho
Escola de Engenharia

Ricardo Alexandre Gonçalves Carotta Martins

Emotional state detection through text analysis

Doctoral Thesis
Doctoral Program in Informatics

Work developed under the supervision of
Paulo Novais
Pedro Henriques

COPYRIGHT AND TERMS OF USE OF THIS WORK BY A THIRD PARTY

This is academic work that can be used by third parties as long as internationally accepted rules and good practices regarding copyright and related rights are respected.

Accordingly, this work may be used under the license provided below.

If the user needs permission to make use of the work under conditions not provided for in the indicated licensing, they should contact the author through the RepositoriUM of Universidade do Minho.

License granted to the users of this work



Creative Commons Atribuição-NãoComercial-Compartilhalgual 4.0 Internacional
CC BY-NC-SA 4.0

<https://creativecommons.org/licenses/by-nc-sa/4.0/deed.pt>

Acknowledgements

First of all, I would like to thank the professors Pedro Henriques, Paulo Novais and José João Almeida for all support, discussions and good times during this work. Also, I would like to thank the ISLab members, for all lunches, soccer, and good ideas that enriched this work.

For my family: Luciana - love of my life, Manuela - my little princess, thank you to embrace one more time another crazy idea of mine. Without you it couldn't be possible !

For my parents and brother, thank you for all positive words and support during difficult times.

At last, thank you for all friends - I will not cite individually here because I would be unfair if forget someone - who cheered for this work. In many times you gave me inspiration to keep in the right path.

"Só levo a certeza de que muito pouco sei." (Almir Sater)

PREVIEW

Resumo

Detecção do estado emocional através de análise de textos

Atualmente, as pessoas são submetidas a rotinas intensas e muitas vezes exaustivas para que possam acomodar todas as expectativas e realizar seus desejos, sejam pessoais ou profissionais. Trabalhadores que exercem suas profissões diurnas e durante a noite tornam-se estudantes universitários, mães que têm que acomodar sua jornada profissional com as tarefas domésticas e jovens que dividem seu tempo em vários estudos escolares e profissionais - muitas vezes tendo que ajudar nos os lares - são alguns exemplos de perfis de pessoas que correm o risco de ter um problema emocional.

Apesar de atingir um grande número de pessoas, não é trivial saber quando alguém está a atingir seu limite, pois é impossível conectar fios e dispositivos que coletem dados por alguns dias para identificar problemas futuros. Além disso, o acesso a esse tipo de equipamento exige dinheiro e tempo, o que não é para todos. A abordagem proposta aqui é coletar dados para inferir o estado emocional atual, tais como (*stress*, fadiga, ansiedade, etc.) de forma não invasiva, transparente, barata, simples de usar e de fácil integração com outros sistemas, através da análise de textos curtos, como a troca de mensagens, redigidos através dos mecanismos de comunicação rápida hoje em voga (chats de e-mails e redes sociais, blogs, fóruns de discussão, SMS, etc.).

Para isso, a ideia é ensinar o computador a identificar as pistas deixadas nas mensagens de texto que revelem o estado emocional do autor. Usando técnicas de aprendizado de máquina (*machine learning*) e mineração de texto, várias mensagens previamente coletadas de diferentes fontes serão analisadas a fim de criar um modelo que classifique o estado emocional. Posteriormente, usando esse modelo de classificação, novas mensagens de texto podem ser analisadas para inferir o estado emocional atual do autor.

Após utilizar diferentes técnicas para extrair as emoções de textos, essas informações foram sintetizadas na criação um perfil emocional que foi utilizado em tarefas de classificação para identificar doenças como depressão, e prever comportamentos tanto individuais como coletivos, atingindo 98% de precisão na detecção de depressão.

Palavras-chave: Análise de Sentimentos, Aprendizado de Máquina, Processamento de Linguagem Natural

Abstract

Emotional state detection through text analysis

Daily a large number of people end up suffering car accidents, heart attacks and emotional problems due to stress. Some reasons like sedentary lifestyle and the poor quality of life experienced currently are easily associated with them in the literature.

Despite reaching a high number of people, it is non-trivial to figure out when someone is in his emotional threshold; especially when connecting body sensors, there is no alternative to collect data. Furthermore, monitoring emotional state conditions continually requires money and time. Our approach to collect human data is the analysis of text messages gathered from email or social networks chats, blogs, SMS's, and other fast communication mechanisms popular at present. This approach here proposed and discussed is useful to measure up the current emotional state (stress, fatigue, etc.) of a person in a non-invasive manner, transparent, cheap, simple to use, and easy to carry around through mobile devices usage.

To achieve this objective, the idea is to *teach* the computer to identify cues left in text messages which reveal the emotional state of the author. By using machine learning and text mining, several messages previously collected from different sources are analysed to create a model which classifies the emotional state. Later, using this classification model, new text messages can be analysed to classify the current emotional state.

After performed different techniques to extract emotions from texts, these information were used to create an emotional profile that was used in classification tasks to identify diseases such as depression, or predict individual or group behaviours, achieving 98% precision in depression detection.

Keywords: Machine Learning, Natural Language Processing, Sentiment Analysis

Contents

List of Figures	xiii
List of Tables	xv
1 Introduction	1
1.1 Motivation	2
1.2 Objective and Research Questions	3
1.3 Document Structure	3
I State of the Art	5
2 Emotions	7
2.1 Discrete Models	8
2.2 Dimensional Models	9
2.3 Cognitive Models	10
2.4 Affective Computing	14
2.5 Detecting Emotions Challenges	14
2.5.1 Detecting Emotions in Texts	15
2.5.2 Detecting Emotions in Speeches	24
2.5.3 Detecting Emotions in Images	25
2.6 Summary	26
3 Human Behaviour	27
3.1 Identification of human personality	28
3.2 Personal lexicon	30
3.3 Summary	31

4	Text Mining and NLP	32
4.1	Applications of Text Mining	33
4.1.1	Information Retrieval	33
4.1.2	Information Extraction	34
4.1.3	Categorization	34
4.1.4	Natural Language Processing	35
4.2	Techniques of Text Preprocessing	36
4.2.1	Tokenization	37
4.2.2	Stemming	38
4.2.3	Lemmatization	40
4.2.4	Part of Speech	41
4.2.5	Word Sense Disambiguation	42
4.2.6	TF-IDF	43
4.2.7	Stopwords	44
4.2.8	N-Grams	44
4.3	Summary	45
5	Word Representation	46
5.1	Word Representation Approaches	46
5.1.1	Dictionary Lookup	46
5.1.2	One Hot Encoding / Bag of Words	47
5.1.3	Word Embeddings	48
5.2	Text similarity	52
5.2.1	Jacquard Coefficient	52
5.2.2	Euclidean Distance	53
5.2.3	Cosine Similarity	54
5.2.4	Word Mover's Distance	56
5.3	Summary	57
II	Application & Architecture	58
6	Emotional State Classifier	60
6.1	Architecture	60
6.1.1	Modules	61
6.1.2	Web API	62
6.1.3	InputDataTraining	63
6.1.4	RequestClassification	64
6.1.5	RequestRegression	66

6.1.6	PredictEmotionalLevels	68
6.2	Tasks	70
6.2.1	Data Preparation	71
6.2.2	Emotional Profiler	72
6.2.3	Vocabulary Personalizer	73
6.2.4	Time Series Personal Model	75
6.2.5	Classification & Regression Model Builder	76
 III Case Studies		 78
7	Case Study 1 - User identification by emotional profile	80
7.1	Data analysis	81
7.2	Polarity analysis	82
7.3	Lexicon-based emotion analysis	84
7.4	Machine learning-based emotion analysis	85
7.5	Conclusion	86
8	Case Study 2 - Lexicon personalization	87
8.1	Related work	87
8.2	Lexicon expansion process	88
8.2.1	Corpus creation	88
8.2.2	Vocabulary creation	88
8.2.3	Similarities	89
8.2.4	Synonyms	90
8.3	Results	90
8.4	Conclusion	92
9	Case Study 3 - Impact of emotions in usual tasks	94
9.1	Theory of Basic Emotions	95
9.2	Related work	96
9.3	Data creation	96
9.3.1	Meta-information	97
9.3.2	Emotional Analysis	97
9.4	Data analysis	98
9.4.1	Emotional correlations	98
9.4.2	Machine learning predictions	99
9.5	Conclusion	101

10 Case study 4 - Determining emotional profiles based in textual analysis	102
10.1 Emotion theories	102
10.2 Related work	103
10.3 Data analysis	104
10.3.1 Preprocessing	105
10.3.2 Polarity analysis	106
10.3.3 Emotional analysis	107
10.3.4 Grammatical analysis	108
10.3.5 Similarity analysis	108
10.4 Conclusion	110
11 Case Study 5 - Prediction of election results according to emotional analysis	111
11.1 Related work	112
11.2 Dataset creation	112
11.2.1 Out of scope	113
11.2.2 Lexicon expansion	113
11.2.3 Preprocessing	115
11.2.4 Sentiment Analysis	116
11.3 Data analysis	116
11.3.1 Training dataset	117
11.3.2 Predicting results	117
11.4 Conclusion	118
12 Case Study 6 - Depression classification based social media analysis	120
12.1 Related work	121
12.2 Identification of depressive profiles	121
12.2.1 Emotional profile	122
12.2.2 Data sample creation	123
12.2.3 Tweets preprocessing	123
12.2.4 Sentiment Analysis	124
12.3 Data analysis	125
12.3.1 Exploratory Analysis	125
12.3.2 Clustering analysis	126
12.3.3 Machine and Deep Learning analysis	127
12.4 Conclusion	128

IV Summary, Contributions and Future Work	129
13 Conclusion	130
13.1 Research questions & Results and contributions	131
13.2 Publications	132
13.3 Future Work	134
Bibliography	135

PREVIEW

List of Figures

1	The six major emotions	8
2	The circumflex model of affect	9
3	The Watson's two-dimensional structure of affect	10
4	Plutchik's wheel of emotions. Extracted from [157]	11
5	OCC Model	12
6	Approaches for Sentiment Analysis in text. Source: author	15
7	WordNet online search	17
8	SentiWordNet Synset's properties. Adapted from https://ontotext.fbk.eu/sentiwn.html	18
9	SentiWordNet records	19
10	Support Vector and Margin identification. Adapted from https://medium.com/mlearning-ai/support-vector-machine-svm-algorithm-a5acaa48fe3a	22
11	Non-linearly separable data. Source: author	22
12	Non-linearly separable data. Source: author	23
13	SVM Multiclass classification process. Source: author	23
14	Text Mining overview. Source: author	32
15	Process of Information Extraction from Texts. Source: author	35
16	Syntactic Tree	36
17	One Hot Encoding / Bag of Words Representation	47
18	Illustration of the Continuous Bag-of-Word (CBOW) and Skip-Gram models. Source: [102]	50
19	CBOW network. Source [53]	50
20	Skip-gram network. Source https://towardsdatascience.com/word2vec-skip-gram-model-part-1-intuition-78614e4d6e0b	51
21	GloVe processing. Adapted from https://www.kdnuggets.com/2018/04/implementing-deep-learning-methods-feature-engineering-text-data-glove.html	52
22	Euclidean Distance. Source https://morioh.com/p/05256ee15f96	54

LIST OF FIGURES

23	Right-Angled Triangle	55
24	WMD Example. Adapted from [94]	56
25	Similarity values. Adapted from [94]	56
26	Model's architecture	61
27	Learning Module	61
28	RequestClassification pipeline	64
29	RequestRegression pipeline	66
30	PredictEmotionalLevels pipeline	68
31	Data Preparation pipeline	71
32	Emotional Profile pipeline	73
33	Emotional Profile pipeline example	73
34	Personal lexicon creation pipeline	74
35	Time Series Personal Model Builder pipeline	76
36	Classification Model module pipeline	77
37	Preprocessing tasks	82
38	Preprocessing text example	83
39	Polarities distribution by category	83
40	Polarities distribution by author	84
41	Lexicon expansion process	89
42	Lexicon polarities	91
43	Proportions	91
44	Dataset creation process	96
45	Preprocessing pipeline	98
46	Preprocessing tasks	105
47	Correlations by author	108
48	Dataset's creation pipeline	113
49	Lexicon expansion process	114
50	Preprocessing tasks	115
51	Plutchik's wheel of emotions	122
52	Preprocessing tasks	124
53	Dataset created	125
54	Outliers identification	126
55	Clusters of information	127

List of Tables

1	Example of synsets	18
2	ANEW values example	20
3	EmoLex lexicon	21
4	Dictionary Lookup example	47
5	Co-occurrence matrix	48
6	Levels of relationship	56
7	EmoLex lexicon snapshot	74
8	Training dataset example	76
9	Posts authors	81
10	Polarities by author	84
11	Basic emotions average per author	85
12	Correlation between polarities and emotions	85
13	Detailed accuracy results for non-preprocessed and preprocessed texts	86
14	Lexicon comparison	90
15	Bill Gates' speech analysis	92
16	Donald Trump speech analysis	92
17	Correlations between <i>AmountErrors</i> and emotions	99
18	Algorithms correlations and their Root Mean Squared Error(RMSE)	100
19	Ranges of <i>AmountErrors</i> per emotions	101
20	Tweets authors	104
21	Polarities analysis from authors and their top 5 audience	106
22	Basic emotions per author	107
23	Basic emotion's frequency average of audience's author	107

LIST OF TABLES

24	Correlation between basic emotion's authors and frequency average basic emotion's top audiences	107
25	Correlation between basic emotion's authors	108
26	Grammatical style for authors and audiences	109
27	Correlation of grammatical frequency average between authors and audience	109
28	Similarities between authors and audiences	110
29	Characteristics of the personal lexicon created	115
30	Algorithms evaluation	118
31	Emotions in second round's day	118
32	Correlations between depressive status and basic emotions	125
33	Correlation between depressive status and intensities	126
34	Benchmark of Machine and Deep Learning algorithms	128

Introduction

Every day, a vast number of people take photos, make videos and send texts using their mobile equipment (smartphones, tablets, etc.). Businesses around the world collect data on consumer preferences, purchases and trends. Supported by appropriate regulations, governments collect all sorts of data concerning traffic over the internet to incident reports in police departments. This amount of data is growing fast. According to [33], in 2020 for each minute of the day, users share 150000 messages in Facebook, Amazon ships 6659 packages, WhatsApp users share 41666667 messages and users apply for 69444 jobs in LinkedIn. And this only a slice of the daily data production. The same company predicted in 2018 that in 2020 every person on earth will produce 1.7 Mb of data each second [32].

This massive amount of data is generated from several sources (and not restricted to): a single person, social groups or companies. For instance, personal smartwatches can measure and store the heartbeat; smartphones can calculate travelled distances, iPods can estimate the number of calories burned during the running, websites can store the user's navigation information. Regarding the data generated by a person and among many other sorts, all texts produced in the context of the so-called Computer Mediate Communications—like blogs, comments as replies to Social Network posts or Newspaper articles, chats, tweets , and so on — can contribute as a source of information characterizing the author.

All this individual/personal generated data, does not reveal, in a primary analysis, relevant information. However, after an in-depth analysis, it can provide relevant knowledge about the person involved.

Using data to infer information about users is not new. Researching on customer buying behaviour over time can reveal some unsuspected patterns – the most famous example is the relation between diapers and beer sold in larger than usual quantities on Friday nights in a retailer. The hypothesis was that husbands had to shop for diapers for the weekend while in their way home from work - and while there, they would pick up beer for the weekend sports on TV. This relationship allowed the store to locate diapers and beers close to each other so that more husbands might be reminded of the one they were most likely to forget.

In science areas like psychology and medicine, the textual information produced by a user can also be an essential source to identify patterns. For [62], expressing emotions can be done through writing, body language, or talking with other people. [48] link negative emotions to increased stress.

The automatic detection of emotions in texts is becoming essential in many different areas such as educational/edutainment games to collect feedback from users, financial to predict stock market prices and healthcare to detect the mood of patients while in treatment. When handling with emotions, there is common thinking that exists a straight connection between emotions, mood and personal profile, and it is a half-truth because mood expresses emotions. However, the mood is different from the emotional profile (or personality) while the emotional profile of a person stays rigid throughout our life, the mood is more open to change, influenced by the current emotions. In addition the emotional state is a snapshot of the mood in a specific moment, being the emotional states' collection along the time similar to the emotional profile.

1.1 Motivation

Emotions are present in most of our everyday occasions and manifestations during our life, such as in decision making and social relations. The research contributions in this area are significant and can provide theoretical and practical advances in human-machine interaction, and a better understanding of the technology influence has on human development [155], allowing the computer to adapt to people and not the opposite. The study of the emotions identification in texts is inserted in a multidisciplinary context, going from Psychology, through the Mining of Texts and Pattern Recognition to Human-Computer Interaction. The research area, known as Opinion Mining, has been experiencing growth, with many researchers working in automatic opinion evaluation on e-commerce and Opinion portals.

The PhD work here discussed aims at presenting a method for the identification of the emotional state through the processing of texts written in English. The emotions to be identified in the texts refer to the emotions proposed in the Plutchik's wheel of emotions [157]. Neutral texts (without emotion) are also identified by the method. For the evaluation of the proposed method, will be constructed a corpus of text messages of people demonstrating different levels of sentiment and a tool that allows the conduction of experiments with different configurations.

The motivation for this work is creating software that analyses the content of messages, extracting the emotions contained into these messages to infer the author's emotional profile and - using this information - classify the emotional state that the author presents at the writing time.

This classifier can be used in many different situations (and not limited to these ones): in a psychological treatment scenario, where a patient needs to be assessed constantly; in the scholarly context, identifying cues about bullying based on the student's emotional profile changing; in the labour context, helping to identify the workers under risk of stress.

However, this follow-up implies constant data collection, which can be considered somewhat invasive. In times when the limits of privacy and how a person's data are treated are discussed, it is necessary that the people whose texts are under analysis are aware and in agreement with this monitoring.

1.2 Objective and Research Questions

The objective of the doctoral project here discussed is to resort to techniques like text mining and machine learning to extract and classify emotional information, from unstructured documents, in order to identify the sentiment contained in the text that can be used to characterise the text's author emotional profile.

Thus, the research questions that will guide the execution of the work are:

- A. Do emotional labels impact on the accuracy of classifications of the emotional states?
- B. Is there an abstract model which differentiate the emotions considering personal perspectives ?
- C. Based on the emotional state of a person, would be possible to predict his actions, choices or some situations which inspire concerning ?

To guide the answers for these questions, it were formulated these hypothesis:

- A. The current emotional state of a person is expressed in his texts when writing;
- B. The set of emotions detected in a large collection of texts of an author during a large interval represent his emotional profile;
- C. The emotional state can be classified according to variations with the emotional profile.

The main contribution of this project will be an emotional state classifier through text analysis, natural language processing and machine learning algorithms.

At a formal description level, the proposal is the creation of an emotional state classifier ES:

$$ES = f(T) \quad (1.1)$$

where:

$T = \{txt_1, txt_2, ..., txt_n\}$ is a set of texts

and

$TXT = \{w_1, w_2, ..., w_n\}$ is a set of words, where w can contain emotional information.

1.3 Document Structure

This thesis is organized in 4 parts and begins with an introduction to the context, motivation and objectives in Chapter 1.

Part I includes Chapters 5 to 2 and presents the State of the art.

Part II just includes one chapter intended to present the PhD proposal.

Part III includes Chapters 7 to 12, aiming at presenting and discussing the case studies to which the proposed emotional classifier was applied.

Part IV presents a summary presented and some conclusions as well as and some directions for future work are presented.

PREVIEW

Part I

State of the Art

This part introduces the state of art of the theories and technologies that were used during the work. First, it is presented abstract concepts and theories from psychology, such as emotions and the different models to represent them, the human personality and the personal vocabulary as source of personality differentiation.

Later, the a set of techniques from Natural Language Processing are presented, in order to explain how the information can be retrieved from texts and modelled to represent the abstract concepts presented earlier.

PREVIEW

Emotions

Although emotions have been studied in several areas of knowledge such as Psychology, Neuroscience, Philosophy and Artificial Intelligence, there is no consensus on the definition of emotion because some issues such as its subjective nature, the divergence of researchers as to its origin, and the term used to describe a wide range of cognitive and physiological states [57]. This lack of consensus reinforces the idea of [47] which claimed that “everyone knows what emotion is until asked to give a definition. Then, it seems no one knows.”

For [96], the concept of emotion can vary among several definitions, depending on the area of knowledge from which they arise. In the psychological and behavioural area, emotions can be understood as systemic responses that occur when highly motivated actions are delayed or inhibited. Thus, emotions concern the execution of something relevant to the organism.

[29] highlighted the fundamental role of emotions in the process of adaptation of living beings. According to the author, a facial expression indicating emotional states is a form of communication that is efficient and understandable to people, regardless of culture, and these striking characteristics are adaptive in all life forms.

In cognitive sciences, according to [28], emotions are body movements or actions visible to third parties. On the other hand, the sentiments are hidden in the organism where they occur, invisible to the public. Sentiments can arise by merely thinking about an event, and imagining what would happen if it would occur.

The differentiation presented above allows noticing that emotions in cognition have a much more visible, as well as conscious, role in the day-to-day. For example, the mood is widely used in educational settings as a way of motivating attention, developing affective sentiments towards the taught content, and promoting a more enjoyable learning experience [140].

Different scientific theories of emotion have been created over the years of research in cognitive sciences, each trying to explain the diversity of affect phenomena. These theories gave rise to three main models of emotions: discrete models, also called categorical ones; dimensional models; and models based on appraisal theory (cognitive models).

2.1 Discrete Models

Discrete model theories propose the existence of basic emotions (happiness, anger, sadness, surprise, disgust, and fear, for instance) that are universally displayed and recognised [38, 88], grouping emotions into categories and assuming that they are independent.

Many researchers have attempted to identify some basic universal emotions which are familiar to all people and differ one from another in significant ways. A famous example is a cross-cultural study by Paul Ekman and his colleagues, in which they concluded that the six basic emotions are anger, disgust, fear, happiness, sadness, and surprise [36]. One of the main advantages of discrete models is that, through psychophysical experiments, the perception of emotions by human beings is discrete. In this way, these models can easily associate emotions with facial expressions that represent them. The six basic emotions of Ekman are represented by six universal facial expressions, since they are understandable by people in different locations, regardless of culture. The facial expressions of the six basic emotions are shown in Figure 1.

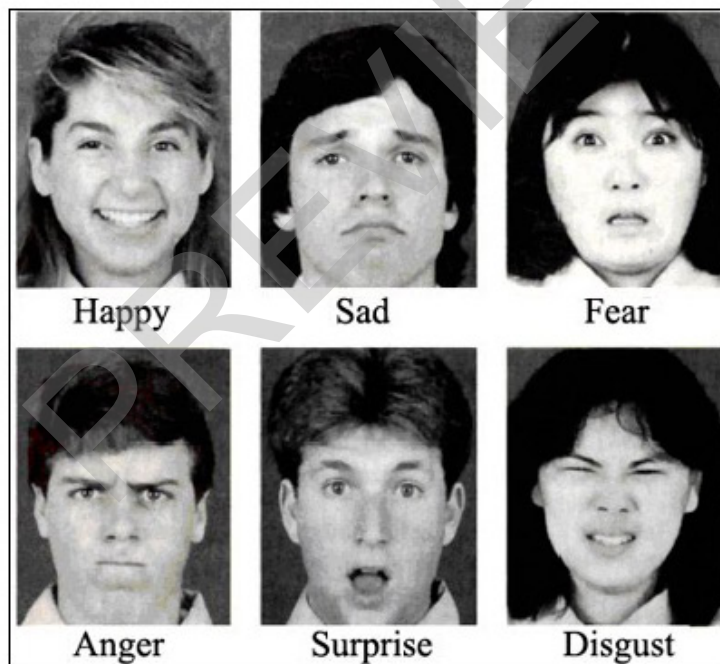


Figure 1: The six major emotions

A significant part of the work on emotion mining and classification from text has been adopted these basic emotion set [75, 160, 187]. For example, to model public mood and emotion, [75] extracted from Twitter six dimensions of mood including tension, depression, anger, vigour, fatigue and confusion. [187] created a significant data set with six basic emotions: anger, disgust, fear, joy, sadness and surprise.

However, there is no consensus on which human emotions should be categorised as basic and be included in the basic emotion set. Moreover, the emotional disambiguation is a contested issue in emotion

research. For instance, it is unclear if "surprise" should be considered an emotion since it can assume negative, neutral or positive valence.

2.2 Dimensional Models

Dimensional theories try to explain emotions regarding two or three dimensions. The most frequent dimensional characterisation of emotions uses two dimensions: arousal (intensity) and valence. Valence is related to positive or negative evaluation and is associated with the feeling state of pleasure (vs displeasure). Arousal reflects the general degree of intensity felt. Low arousal is associated with less energy and high arousal with more energy. However, using this two-dimensional is challenging to differentiate emotions that share the same values of valence and arousal, as *anger* and *fear*. For this reason, a third dimension (e.g. intensity) is often added. According to [101] "the third view emphasises the distinct component of emotions, and is often termed the componential view."

Among the dimensional models in 2D, one of the most known is the circumplex model of affect, proposed by [169], in which the emotions are related to each other, organised in a circle and represented in two dimensions: valence (pleasant versus unpleasant sentiment) and activation (awake versus sleepy). Followers of this affective model suggest that each affective experience is a consequence of a linear combination of these two dimensions, resulting in a particular emotion, as presented in Figure 2. Fear, for example, is conceptualised as a neurophysiological state involving the combination of negative valence and increased activation in the central nervous system.



Figure 2: The circumflex model of affect

In his study, [169] presented a list of 28 words that people use to describe sentiment, moods and emotions.